

MATH 189 HW 1

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Question 1

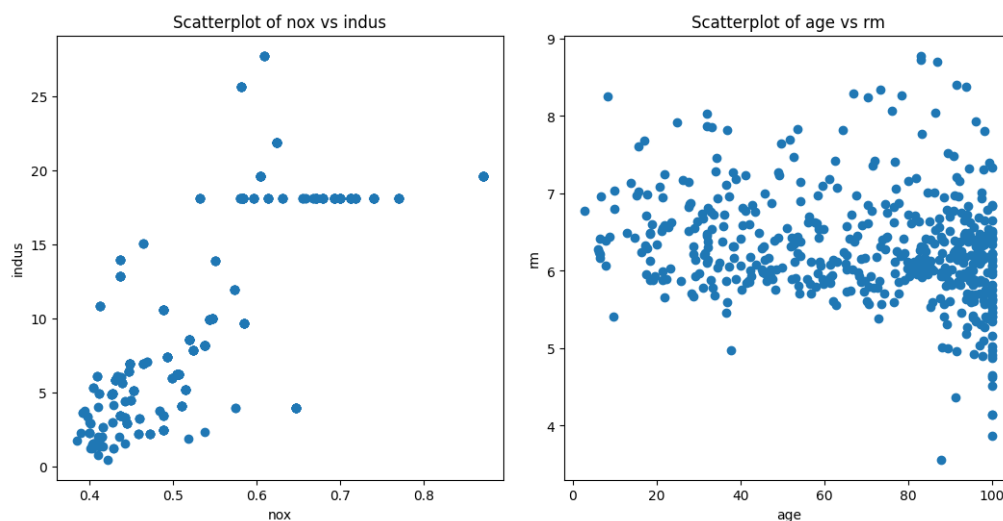
- a. Regression, $n = 500$, $p = 3$
- b. Categorical, $n = 20$, $p = 14$
- c. Regression, $n = 52$, $p = 3$

Question 2

- a. Obs. 1: 2
Obs. 2: 2
Obs. 3: $\sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.16$
Obs. 4: $\sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.23$
Obs. 5: $\sqrt{(-1)^2 + 1^2} = \sqrt{2} \approx 1.41$
Obs. 6: $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \approx 1.73$
- b. Our prediction for $K = 1$ is Green, because that's the color of the closest neighbor.
- c. Our prediction for $K = 3$ is Red, because it is close to two red neighbors and one green neighbors, so by plurality, red is the color of our prediction.
- d. If the Bayes decision boundary is highly nonlinear, then a low K value would provide the closest KNN decision boundary to Bayes. This is because low K values offer higher flexibility in the KNN decision boundary, which better fits a highly nonlinear Bayes decision boundary.

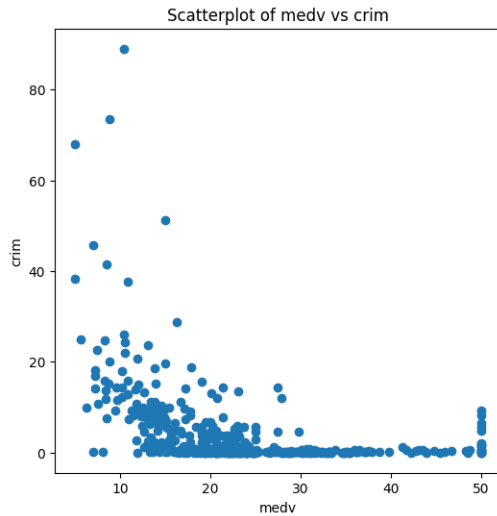
Question 3

- b. 506 rows and 13 columns. Rows represent each data point, columns represent the entries for each data point.



c.

For the first graph, I plotted nitrous oxides concentration versus proportion of non-retail business acres per town. There is a positive correlation between the two, although not a strong one, it also shows heteroscedasticity. For the second graph, I plotted average number of rooms per dwelling versus proportion of owner occupied units built prior to 1940. There seems to be no correlation for these predictors.



d.

For the this graph, I plotted median value of owner-occupied homes versus per capita crime rate by town. There is a weak negative correlation between the two, although linear regression might not be appropriate here.

- e. The highest crime rate per capita is 88.9762 for town index 380, its range is 88.96988 which is a very wide range between the safest town versus the most dangerous town. The highest full-value property-tax rate is \$7,110,000 for town index 488, the range is \$5,240,000, which is again a very wide range between the areas with the highest property tax value and lowest property tax value. The highest pupil-teacher ratio is 22 for town index 354, and the range is 9.4, a moderate sized range.
- f. 35 suburbs in this data set bound the Charles river.
- g. The median pupil-teacher ratio among the towns in this data set is 19.05.
- h. Town index 398 has the lowest median value of owner occupied homes. The values of other predictors are 38.3518 for crime rate 0 for proportion of residential land zoned for lots over 25,000 sq.ft, 18.1 for proportion of non-retail business acres, does not bound the Charles river, 0.6930 for nitrogen oxides concentration, 5.453 for average number of rooms per dwelling. 100 for proportion of owner-occupied units built prior to 1940, 1.4896 for weighted mean of distances to five Boston employment centers, 24 for index of accessibility to radial highways, 666 for full-value property

tax rate per \$10,000, 20.2 for pupil-teacher ratio, 30.59 for lower status of population.

- i. 64 of the suburbs average more than seven rooms per dwelling. 13 of the suburbs has more than eight rooms per dwelling. These suburbs tend to have very low full-value property tax rate, crime rate per capita, and very high median value of owner-occupied homes and proportion of owner-occupied units built prior to 1940.